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TOPICAL REVIEW

Next Generation Human Action Recognition: A Comprehensive Review of State-of-the-Art Signal Processing Techniques

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ABSTRACT Human Action Recognition is a rapidly evolving field at the intersection of computational intelligence, signal processing, and machine learning, driving innovations in healthcare, fitness, surveillance, and smart environments. This review bridges the gap between classical techniques, such as Fourier Transform (FT), Wavelet Transform (WT), and Principal Component Analysis (PCA), and advanced Deep Learning (DL) approaches, including Convolutional Neural Network (CNN), Recurrent Neural Network (RNN) and emerging Transformer-based architectures. A key focus is the integration of multimodal sensing modalities, including accelerometers, gyroscopes, Electromyography (EMG), and electroencephalography (EEG), in conjunction with noncontact technologies such as thermal and infrared imaging, enhancing HAR adaptability in diverse environments. Unlike prior surveys, this work presents a comprehensive synthesis of HAR methodologies, systematically analyzing 217 studies to quantify trends, assess challenges such as noise resilience, real-time constraints, and scalability, and highlight innovative solutions, including advanced multimodal fusion frameworks and energy-efficient architectures. Furthermore, a research roadmap is proposed to advance scalable, real-time HAR systems with deeper multimodal data integration and lightweight computational models, paving the way for robust and adaptable solutions in real-world applications.

INDEX TERMS Human action recognition, vision transformers, multimodal sensor fusion.

I. INTRODUCTION

Human Action Recognition (Human Action Recognition (HAR)) has emerged as a transformative field at the intersection of computational intelligence, signal processing, and machine learning, enabling diverse applications such as healthcare, surveillance, fitness tracking, and Human-Computer Interaction (HCI) [1], [7]. HAR systems rely on multimodal data streams from wearable devices, vision-based systems, and environmental sensors to interpret human actions in dynamic environments with high robustness and accuracy. Despite substantial advances, critical challenges persist—particularly in real-time constraints, scalability, noise resilience, and energy efficiency.

Over time, the HAR methodologies have evolved from traditional signal processing techniques such as FT and

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WT to modern DL architectures such as CNN and RNN. These advancements have significantly improved feature learning and adaptability in complex scenarios [9], [10], [30]. However, while DL techniques enhance performance, they also introduce new challenges related to computational complexity, real-time processing, and resource constraints.

Although there are numerous review studies on the methodologies HAR, key gaps remain. Most reviews focus exclusively on vision-based or sensor-based approaches, often overlook the synergistic potential of multimodal fusion, predictive analytics, and real-time optimization [52], [55]. Although some recent studies explore advanced DL techniques, emerging technologies such as transformer-based models and IoT-driven architectures remain underexplored in the context of HAR, despite their potential to enhance scalability and computational efficiency.

In addition, practical deployment challenges—including handling noisy environments, ensuring real-time responsiveness, and developing energy-efficient artificial intelligence (AI) models—continue to hinder widespread adoption. Addressing these issues requires an integrated perspective that combines signal processing, machine learning, and multimodal sensor fusion to create scalable, real-world HAR applications. Ubiquitous computing devices, including wearables and ambient sensors, enable high-precision action classification by leveraging multimodal data sources. Roggen et al. [71] demonstrated that complex HAR tasks can be significantly improved through multi-sensor integration and advanced temporal modeling.

Beyond technical challenges, HAR systems hold immense potential for healthcare, smart cities, and HCI. In healthcare, HAR plays a vital role in the detection of falls, chronic disease monitoring, and rehabilitation support. In smart cities, HAR-driven systems contribute to public safety and urban planning through real-time surveillance and crowd behavior analysis, as illustrated in Fig. 1.

This review aims to bridge existing gaps by synthesizing insights in signal processing techniques, advanced DL models, and multimodal HAR systems. Table 1 provides a comparative overview of existing HAR surveys, highlighting their primary focus areas and the additional contributions of this study.

A. MOTIVATIONS AND CONTRIBUTIONS OF THIS REVIEW

To address these gaps, this review synthesizes insights in signal processing, machine learning, and multimodal sensor fusion to provide a comprehensive and forward-looking analysis of HAR. The primary contributions of this study are as follows:

- **Bridging Signal Processing & Machine Learning:** Investigating how classical feature extraction methods complement modern DL architectures.
- **Multimodal Fusion for HAR:** Analyzing techniques integrating vision-based, sensor-based, and IoT-driven HAR systems.

TABLE 1. Comparative analysis of human action recognition (HAR) surveys and the proposed study, emphasizing signal processing (SP), machine learning (ML), multimodal fusion (MF), real-time processing (RP), internet of things (IoT), and healthcare applications (Health).

Prior Reviews	SP	ML	MF	RP	IoT	Health
[55]	✓	×	×	×	×	×
[52]	✓	✓	×	×	×	✓
[30]	✓	✓	✓	×	✓	×
[46]	✓	✓	✓	✓	✓	×
[30]	✓	✓	✓	×	×	×
[185]	✓	✓	✓	×	×	×
[186]	✓	✓	✓	✓	✓	×
[187]	✓	✓	✓	×	×	×
[188]	✓	✓	✓	×	×	×
[190]	✓	✓	✓	×	×	×
[207]	×	✓	×	×	×	×
Proposed Study	✓	✓	✓	✓	✓	✓

- **Scalability, Noise Resilience & Real-Time Deployment:** Identifying key challenges and proposing mitigation strategies for practical implementation.
- **Future Research Roadmap:** Highlighting transformers, edge computing, federated learning, and next-generation HAR architectures as potential breakthroughs.

While previous HAR surveys have extensively covered CNN, RNN, and standard datasets, this review adopts a broader system-level perspective. We analyze HAR from foundational signal processing to deployment-oriented challenges, incorporating underexplored but emerging directions such as cognitive radar for contactless sensing, ultra-low power wearable technologies, and transformer-based architectures for real-time inference. Furthermore, our structured comparison tables and hierarchical taxonomy provide a reusable analytical framework for aligning HAR with edge computing, federated learning, and neuromorphic systems. This layered treatment bridges the gap between algorithmic development and scalable, real-world deployment.

The remainder of this paper is organized as follows: Section II details the systematic review methodology. Section III presents the historical progression of HAR, from classical signal processing to advances in DL and the integration of 5G / 6G, cognitive radar and edge computing. Section IV reviews state-of-the-art machine learning and feature extraction techniques. Section V establishes the mathematical framework for HAR, including feature extraction, multimodal fusion, and classification models. Section VI explores various HAR sensor modalities and their roles in recognition tasks. Section VII evaluates existing datasets and discusses benchmarking challenges. Section VIII identifies key bottlenecks, including privacy concerns, sensor

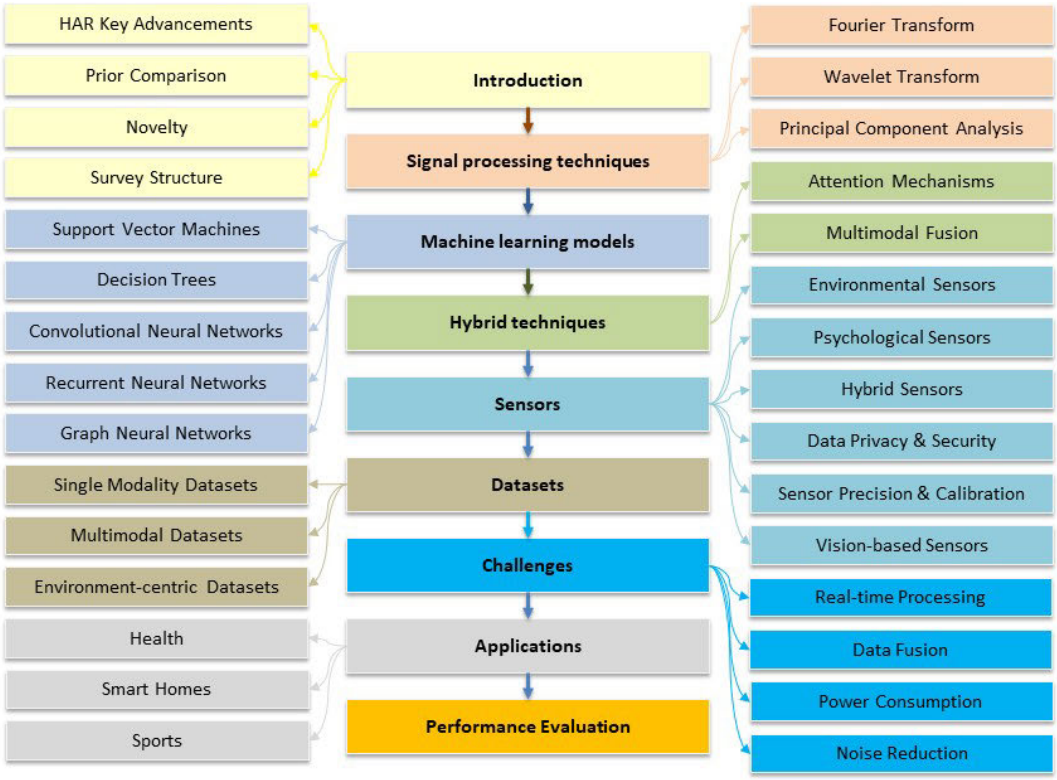

FIGURE 1. Conceptual roadmap of the surveyed HAR literature, illustrating the organization of key themes including signal processing techniques, machine learning models, sensor modalities, datasets, challenges, applications, and performance evaluation.

calibration, and real-time constraints. Section IX explores diverse HAR applications. Section X discusses evaluation metrics for system benchmarking. Finally, Section XI concludes with insights into future research and strategies for scalable, efficient HAR systems.

This review integrates knowledge across multiple disciplines to bridge traditional HAR methodologies and next-generation AI-driven systems, with strong emphasis on real-world deployment and scalability.

II. MATERIALS AND METHODS

This systematic review synthesizes advancements in HAR by integrating insights across signal processing, machine learning, and multimodal sensor fusion domains. A structured search strategy was used to query the major academic databases, including *IEEE Xplore*, *SpringerLink*, *ACM Digital Library*, *Google Scholar*, and *ScienceDirect*, covering publications from 2000 to 2024. Search queries combined keywords related to HAR techniques, signal processing methods, sensor fusion, and DL applications.

Inclusion criteria required studies in English that feature empirical evaluations of methodologies HAR, with emphasis on signal processing or multimodal sensor fusion. Exclusion criteria filtered out purely theoretical works, non-English papers, and duplicates.

The article selection process adheres to the PRISMA guidelines, as illustrated in Fig. 2, describing the stages of record identification, relevance screening, full text eligibility evaluations and final inclusion.

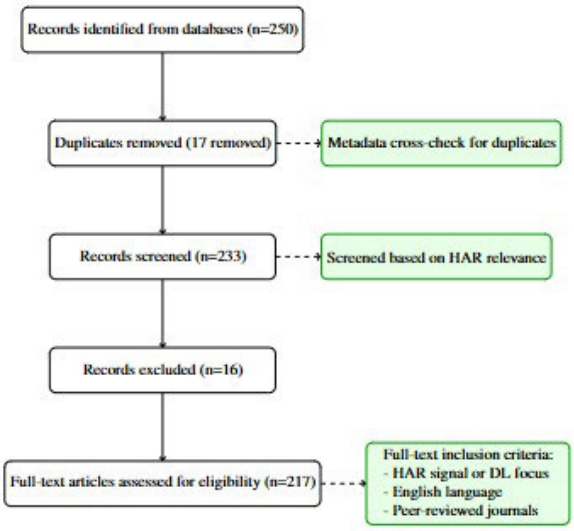

FIGURE 2. Systematic review workflow following PRISMA 2020 guidelines, detailing study identification, screening, eligibility assessment, and final inclusion for HAR literature.

To contextualize the selected studies within the broader landscape of HAR research, the following section traces the historical evolution of key techniques, from classical signal processing to the emergence of deep learning architectures.

III. BACKGROUND AND EVOLUTION

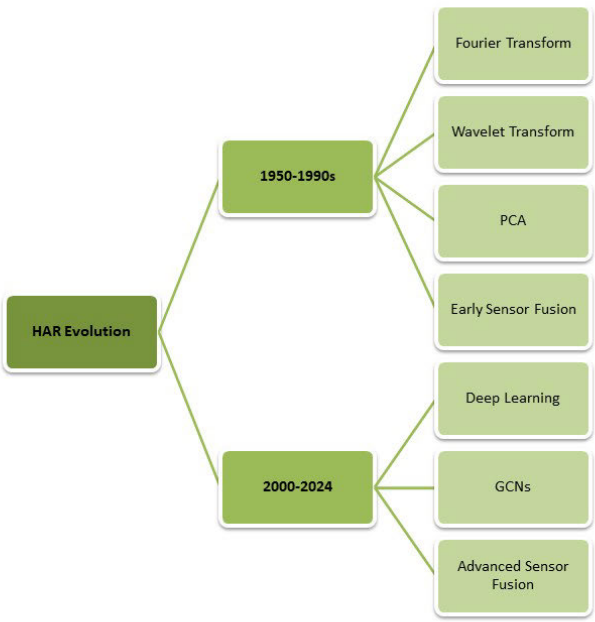

FIGURE 3. Chronological evolution of HAR techniques, from foundational signal processing methods (1950–1990s) to advanced deep learning and sensor fusion frameworks (2000–2024).

Evolution of HAR Techniques: Over the decades, the field of HAR has witnessed substantial advancements in methods and tools. In the 1950s, fundamental techniques such as FT enabled feature extraction from sensor data. By the 1960s, WT improved time-frequency analysis, facilitating more efficient processing of nonstationary signals. In the 1970s, PCA emerged as a powerful technique for feature extraction and dimensionality reduction.

In the 1990s, machine learning methods such as Support Vector Machines (SVMs) and Hidden Markov Models (HMMs), combined with early sensor fusion systems, improved recognition performance in controlled environments. The 2000s marked a turning point with the introduction of CNN and RNN architectures, which enabled automatic feature learning and temporal modeling, significantly enhancing recognition accuracy.

During the 2010s, Long Short-Term Memory (LSTM) networks and Graph Convolutional Networks (Graph Convolutional Network (GCN)) pushed the field forward by modeling sequential and structured data. In parallel, novel sensing paradigms, such as cognitive radar and Wi-Fi-based localization, began gaining traction. The integration of 5G/6G connectivity between 2020 and 2023 enabled real-time, scalable, and cloud-integrated HAR systems.

Looking ahead, 2024 and beyond signal the rise of Transformer-based architectures and advanced sensor fusion frameworks, which are expected to set new benchmarks for accuracy, efficiency, and cross-domain applicability.

This chronological advancement underscores the interdisciplinary nature of HAR, where innovations in signal processing, machine learning, and embedded sensor technologies converge to meet the growing complexity of real-world deployment. Fig. 3 summarizes the evolution of HAR methodologies—from foundational signal processing to emerging DL paradigms.

This historical trajectory lays the foundation for the following section, which details the core signal processing and machine learning frameworks that enable robust action recognition in modern HAR systems.

IV. TECHNOLOGICAL FOUNDATIONS

Human Action Recognition (HAR) systems have undergone transformative advancements driven by innovations in signal processing and the proliferation of Internet of Things (IoT) technologies [41], [59], [93]. These systems leverage wearable devices, wristbands, and sensor networks to enable real-time motion analysis using advanced algorithms and communication components. By integrating multimodal data streams with sophisticated machine learning methodologies, HAR systems effectively capture and categorize human actions in diverse domains, including healthcare, smart homes, and urban environments.

A. SIGNAL PROCESSING FOUNDATIONS

Signal processing plays a foundational role in HAR by transforming noisy, high-dimensional sensor inputs into compact, discriminative features suitable for learning. Real-world sensor data is often affected by motion artifacts, environmental interference, and signal drift, requiring robust preprocessing techniques.

Three widely used methods—**Fourier Transform (FT)**, **Wavelet Transform (WT)**, and **Principal Component Analysis (PCA)**—offer complementary benefits in HAR pipelines. These techniques, although classical, remain actively embedded in modern deep learning workflows for feature enhancement, denoising, and dimensionality reduction.

1) FOURIER TRANSFORM (FT)

The Discrete FT (DFT) decomposes time-domain signals into their frequency components:

$$X_k = \sum_{n=0}^{N-1} x_n e^{-j\frac{2\pi}{N}kn}, \tag{1}$$

where X_k is the frequency spectrum of the input signal x_n. Fast Fourier Transform (FFT) variants [4] reduce computational latency, making FT suitable for real-time preprocessing in HAR.

2) WAVELET TRANSFORM (WT)

Unlike FT, the Continuous WT (CWT) provides localized time-frequency decomposition, ideal for non-stationary gestures such as transitions or falls [3]:

$$W(a, b) = \int_{-\infty}^{\infty} x(t)\psi^* \left(\frac{t-b}{a}\right) dt, \tag{2}$$

where ψ is the mother wavelet, a the scale, and b the translation factor.

3) PRINCIPAL COMPONENT ANALYSIS (PCA)

PCA reduces dimensionality by retaining the principal axes of variance [2]:

$$C = \frac{1}{N-1} \sum_{i=1}^{N} (x_i - \bar{x})(x_i - \bar{x})^T, \tag{3}$$

where x_i is the data vector and $\bar{x}$ its mean. PCA is often used in resource-constrained deployments for lightweight encoding.

Each method has trade-offs: FT is best for cyclic signals; WT excels in transient gesture analysis; PCA is ideal for low-power systems. These classical techniques remain highly relevant and are often embedded in deep learning (DL) pipelines to improve saliency and reduce burden in real-world HAR deployments.

B. SIGNAL PROCESSING WITH DEEP LEARNING

Modern HAR systems integrate sensor modalities with DL models like CNNs for multimodal fusion [173]. SP methods (FT, WT, PCA) serve as preprocessing blocks that denoise, enhance features, and reduce dimensionality before feeding into DL pipelines.

- **FT:** Emphasizes periodic motion features.
- **WT:** Highlights localized signal transitions.
- **PCA:** Compresses high-dimensional features while preserving variance.

Chen et al. [108] proposed a hybrid CNN-RNN pipeline that models spatial-temporal features:

$$\begin{aligned} F &= \phi(\mathbf{X}), \\ y &= f(F; \mathbf{W}), \end{aligned} \tag{4}$$

NeuroHar [47] optimizes network structure using genetic encoding:

$$C_i = [H_i, N_i, \alpha_i, \lambda_{1i}, \lambda_{2i}, \eta_i] \tag{5}$$

$$\text{Fitness}(C_i) = \frac{1}{\sqrt{\frac{1}{n}\sum_{j=1}^{n}(y_j - \hat{y}_j)^2}} \tag{6}$$

These methods support the move toward self-evolving, low-cost HAR models capable of adapting to sensor variability and noise.

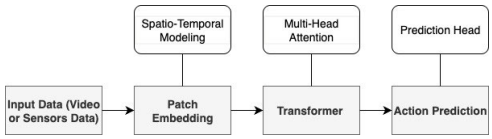

FIGURE 4. Vision Transformer (ViT) architecture for HAR, illustrating patch embedding, transformer-based spatio-temporal modeling, multi-head attention, and final action prediction layers.

C. GRAPH CONVOLUTIONAL NEURAL NETWORKS (GCNNS)

GCNNs extend CNNs to structured data such as skeletal joints. The core graph convolution is:

$$H^{(l+1)} = \sigma\left(D^{-\frac{1}{2}} A D^{-\frac{1}{2}} H^{(l)} W^{(l)}\right), \tag{7}$$

Recent work [217] shows attention-weighted GCNNs improve recognition of complex multi-person gestures in HAR tasks by adaptively learning edge weights.

D. ADVANCED SENSOR FUSION

Sensor fusion techniques combine modalities (e.g., vision + inertial). Fusion can be:

- **Early:** Combine raw signals.
- **Late:** Combine model outputs.
- **Hybrid:** Merge intermediate features.

Nie et al. [169] showed that LoRa + inertial hybrid fusion outperforms single-modality setups. Adaptive fusion via attention weights [14]:

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^{N} \exp(e_j)}, \tag{8}$$

Contactless fusion via WiFi and radar is privacy-preserving [41], [135], [178]. Reinforcement learning and GNNs also guide adaptive re-weighting based on context.

1) VISION TRANSFORMERS (ViTs)

ViTs capture long-range dependencies and are effective for ambiguous actions [214], [215], [216]. Fig. 4 shows a ViT-based HAR pipeline.

E. COMPARATIVE INSIGHTS

The table 2 and 3 synthesize the capabilities and trade-offs of modern HAR technologies (SP, DL, sensor fusion, transformers).

These insights reveal that the deployment context dictates the ideal pipeline: signal processing for interpretability and lightweight setups; deep learning for robust generalization; and transformers/fusion frameworks for complex environments. The next section formalizes these insights mathematically.

V. MATHEMATICAL FRAMEWORK FOR HAR SYSTEMS

Building on the outlined technological bases, an optimized mathematical structure is as following: a combination of advanced signal processing, cutting-edge machine learning

TABLE 2. Summary of HAR categories (C), technologies, and their deployment suitability.

C	Technology	Real-Time	Applications	Challenges	Suitability Context
SP	FT, WT, PCA	✓	Preprocessing	Noise/artifact handling	Embedded HAR
DL	CNN-RNN, GCN, Attention	✓	Spatial-temporal HAR	Training cost	Accurate in high-data scenarios
SF	Hybrid Fusion, RL-based Fusion	✓	IoT/Edge HAR	Sensor sync issues	Robust in dynamic settings
ET	WiFi, Radar, 5G/6G	✓	Contactless monitoring	Signal variability	Privacy-first HAR

TABLE 3. Comparative analysis of state-of-the-art HAR studies across domains.

Study	Methodology	Dataset	Accuracy (%)	Applicability	Use Case
[41]	WiFi Signals	UCI HAR	92.5	High	Smart homes
[108]	CNN-RNN	WISDM, PAMAP2	93.5	High	Mobile health
[217]	Attention-GCNNs	NTU RGB+D	97.8	High	3D HAR (rehab)
[169]	LoRa Fusion	Custom IoT	93.8	High	Rural IoT
[189]	5G/6G Fusion	CASAS	96.0	High	Smart city

frameworks, and progressive innovations to tackle challenges related to real-time processing, multimodal sensor amalgamation, and extensibility.

A. FEATURE EXTRACTION AND REPRESENTATION

Accelerometer-based HAR has been widely adopted for motion classification and physical action tracking. Mobile phones and wearable devices utilize embedded accelerometers to capture raw motion signals, which are subsequently processed to extract meaningful features for Action Recognition (AR). Early implementations, such as Yang [68], demonstrated the feasibility of lightweight HAR by using simple acceleration features on mobile phones, eliminating the need for complex feature engineering. The effectiveness of a HAR system is heavily based on extracting discriminative features from raw sensor data, a process that can be mathematically expressed as (see Equation 9):

$$\mathbf{F} = \phi(\mathbf{X}), \tag{9}$$

where $\mathbf{X}$ represents the unprocessed input data (e.g., sensor signals or video frames), ϕ denotes the feature extraction function, and $\mathbf{F}$ is the resulting feature vector encapsulating critical information from the input.

Traditional methods such as FT and WT, previously introduced in Section IV-A, extract insights from frequency and time-frequency domains, respectively. Modern DL approaches, including CNN, autonomously learn hierarchical representations directly from raw data, removing the need for handcrafted features.

To enhance the robustness of the features and mitigate noise, pre-processing techniques such as Kalman filtering can be integrated into the extraction function ϕ, ensuring that the extracted features remain reliable under varying conditions.

Furthermore, Thakur and Biswas [208] introduced a guided regularized random forest feature selection technique for smartphones HAR, demonstrating its ability to refine the representations of sensor data by selecting the most informative features. Their approach highlights the importance of feature selection in optimizing performance HAR, particularly in resource-constrained environments.

While deep learning-based feature extraction offers superior accuracy and flexibility in large-scale deployments, it often incurs higher energy costs and requires extensive labeled data. In contrast, traditional hand-crafted features (e.g., PCA, FT, WT) remain valuable in edge computing scenarios, where model interpretability, low latency, and computational frugality are prioritized. Thus, a feature extraction strategy should be selected based on deployment constraints - with deep models favored for high-stakes classification and handcrafted techniques for lightweight, real-time, or privacy-sensitive applications.

B. MULTIMODAL DATA FUSION AND ACTION CLASSIFICATION

To enhance recognition accuracy, modern HAR systems increasingly rely on multimodal data fusion. By integrating complementary data from accelerometers, gyroscopes, cameras, and physiological sensors, models capture richer spatial-temporal features for complex action classification. The fusion process can be mathematically described as:

$$\mathbf{F}_{\text{fusion}} = \text{Fusion}(\phi_1(\mathbf{X}_1), \phi_2(\mathbf{X}_2), \ldots, \phi_k(\mathbf{X}_k)), \tag{10}$$

where $\mathbf{X}_i$ represents the input from the i-th sensor, and ϕ_i denotes its respective feature extraction function. Fusion strategies include concatenation, weighted averaging, and attention-based methods.

To manage sensor variability in IoT environments, latency is a critical factor:

$$\text{Latency}_{\text{fusion}} = \sum_{i=1}^{k} \frac{\text{Processing Time}(\mathbf{X}_i)}{\text{Bandwidth}}, \tag{11}$$

Efficient communication and compression protocols are required to meet real-time constraints.

The final classification stage maps fused features to action labels using a model f with parameters $\mathbf{W}$:

$$\hat{y} = f(\mathbf{F}; \mathbf{W}), \tag{12}$$

Classifiers such as CNN and RNN Transformers, and ensemble models handle temporal dependencies and spatial hierarchies effectively. Performance efficiency is particularly critical in edge computing:

$$\text{Energy Efficiency} = \frac{\text{Accuracy}}{\text{Computational Cost}}, \tag{13}$$

C. OPTIMIZATION FOR REAL-TIME HAR

To reduce computational overhead and meet deployment constraints, optimization strategies such as pruning, quantization, and knowledge distillation are employed. The overall latency in HAR system can be decomposed as:

$$\begin{aligned}\text{Total Latency} &= \text{Data Acquisition Time} \\ &\quad + \text{Feature Extraction Time} \\ &\quad + \text{Classification Time}.\end{aligned} \tag{14}$$

These components can be minimized through lightweight model design and hardware-aware optimization.

D. PROPOSED INTEGRATIVE FRAMEWORK FOR SCALABLE HAR

Based on the above principles, we propose an integrative framework for real-time, scalable HAR:

1) **Fusion Layer:** Applies attention-based weighted fusion:

$$\mathbf{F}_{\text{weighted}} = \sum_{i=1}^{k} w_i \cdot \phi_i(\mathbf{X}_i), \tag{15}$$

where w_i is contextually adaptive for each sensor input.

2) **Adaptive Classifier:** Leverages ensemble learning to improve generalization.

$$\hat{y} = \text{MajorityVote}\big(f_1(\mathbf{F}), f_2(\mathbf{F}), \ldots, f_n(\mathbf{F})\big). \tag{16}$$

3) **Scalability Module:** Employs MobileNet and edge-aware deployment strategies to enable low-power processing on wearable and smartphone platforms.

Köping et al. [125] proposed a general framework for HAR pipelines across sensor types. Our approach extends this with latency-aware fusion and adaptive learning components for noisy, real-world conditions. Smartphones, in particular, offer a scalable deployment due to ubiquitous motion sensors and wireless capabilities. San-Segundo et al. [210] demonstrated the efficacy of HMMs in modeling temporal patterns on mobile platforms.

While Section V established the mathematical backbone of HAR systems, their real-world performance is fundamentally constrained by the sensors that generate the raw input. The next section categorizes the primary sensor modalities used in HAR to close this abstraction gap, highlighting their respective roles, advantages, and deployment contexts.

VI. SENSOR TAXONOMY

HAR systems rely on diverse sensors to capture the multimodal data necessary for robust AR. Figure 5 provides an overview of the main sensor categories: wearable, vision-based, physiological, environmental, and hybrid sensors.

A. TYPES OF SENSORS AND THEIR ROLES

1) WEARABLE SENSORS

Devices such as accelerometers, gyroscopes, and magnetometers capture motion and physiological signals, enabling fitness monitoring and health diagnostics [21], [23], [58], [69], [117], [129], [132], [194]. Strategic placement (e.g., wrist, ankle, chest) and miniaturization enhance accuracy and user comfort.

2) VISION-BASED SENSORS

Cameras and RGB-D sensors provide rich spatial information for AR, excelling in controlled environments but facing challenges with occlusion and privacy [102], [190], [193].

3) PHYSIOLOGICAL SENSORS

EMG, ECG, and textile pressure sensors capture muscle and heart action, supporting applications in rehabilitation and health monitoring [22], [144], [146].

4) ENVIRONMENTAL SENSORS

Microphones, pressure, and temperature sensors monitor environmental changes, enabling context-aware HAR in smart homes and ambient intelligence [142].

5) HYBRID AND AMBIENT SENSORS

Hybrid systems combine multiple modalities (e.g., inertial + RF, capacitive sensing) for enhanced robustness and non-intrusive monitoring [26], [44], [45], [154], [155], [156], [157], [159].

For a direct comparison of sensor-based and vision-based HAR challenges, see Table 4.

Table 4 summarizes the contextual trade-offs between sensor-based and vision-based HAR systems. Sensor-based methods are generally more energy-efficient and privacy-preserving, while vision-based systems provide richer spatial information but at higher computational and privacy costs.

TABLE 4. Comparison of key challenges between sensor-based and vision-based HAR systems across real-world deployment factors.

Challenge	**Sensor-Based HAR** (Wearables, IoT)	**Vision-Based HAR** (RGB, Depth, Thermal)
Data Collection	Needs user compliance; sensor drift risk	Sensitive to occlusion; camera setup affects results
Real-Time Processing	Low latency; lightweight inference possible	High compute cost; hard to achieve real-time speed
Occlusion & Coverage	Affected by placement and motion artifacts	Vulnerable to full/partial occlusion; view-dependent
Lighting Conditions	Not affected by illumination changes	Accuracy drops in poor lighting or glare
Privacy Concerns	User discomfort with wearables in some cases	High surveillance risk; raises ethical concerns
Scalability	Needs user/device calibration	Storage-heavy; limited scalability in multi-user setups
Energy Efficiency	Supports long-term use with low power	Consumes significant energy for high-res streams
Environmental Adaptability	Flexible in IoT-enhanced deployments	Struggles in dynamic/uncontrolled conditions
Cost	Affordable; basic sensors are cheap to deploy	High-cost cameras and compute infrastructure
Overall Trade-offs	Cost-effective, but needs calibration	High data quality, but resource-intensive

B. MULTIMODAL SENSOR INTEGRATION

Modern HAR systems increasingly leverage multimodal sensor integration to enhance robustness, accuracy, and adaptability. For a review of fusion paradigms (early, late, hybrid, adaptive), see Section IV-D. Notable advances include:

- Depth camera + inertial fusion for complex AR [111].
- GCN-based multimodal fusion for spatial-temporal learning [188].
- Robust frameworks for wearable, ambient, and physiological sensor fusion [12], [54], [79], [113].

Indoor positioning (Wi-Fi, Radio-Frequency Identification (RFID), Ultra-Wideband (UWB)) augments HAR by providing spatial context [100]. Mobile and IoT-based HAR enables real-time tracking, but faces challenges in user variability and device placement [64].

Recent advances in DL and attention mechanisms for multimodal fusion are detailed in Section IV-B.

The selection of a fusion strategy must align with deployment constraints and modality characteristics. **Early fusion** is lightweight and suitable for low-latency systems, but its performance may degrade when sensor streams are misaligned or asynchronous. **Late fusion** provides modularity and fault tolerance, making it ideal for heterogeneous or partially missing data scenarios. **Hybrid fusion** balances these trade-offs and is preferred in systems combining wearable and ambient sensors. Finally, **adaptive fusion**—often guided by attention or learning-based policies—is best suited for personalized or dynamic HAR settings, such as elderly care, where sensor relevance evolves.

C. SENSOR PLACEMENT AND SIGNAL PROCESSING

Sensor placement critically impacts HAR accuracy. Strategic positioning (e.g., on wrist, ankle, or chest) and calibration improve generalization and reduce signal distortion [30], [70], [206]. Device-free HAR (e.g., ambient, RF-based) eliminates the need for worn sensors but requires careful environmental integration [43], [75].

For mathematical details and signal processing techniques (FT, WT, PCA), refer to Section IV-A.

These studies are summarized in Table 5 as following:

Table 5 highlights that no single sensor type universally outperforms others across all HAR applications. Wearable sensors are ideal for continuous motion tracking and energy efficiency, especially in personal health monitoring. In contrast, vision-based sensors offer rich spatial context but are constrained by lighting conditions and privacy concerns. Physiological sensors provide internal state information, making them valuable in the clinical and affective computing domains, although they may require invasive setups. Environmental sensors are unobtrusive and suitable for smart homes, but can lack specificity. Hybrid sensors offer the best of both worlds by combining modalities, but their deployment is often complex and costly. Thus, sensor selection should align with application constraints such as privacy, energy budget, accuracy needs, and environmental conditions.

TABLE 5. Overview of sensor types used in HAR, with representative studies and key application contexts.

Sensor Type	Representative Study	Core Contributions and Use Cases
Wearable Sensors	[58]	Leverage accelerometers, gyroscopes, and IMUs for fine-grained motion tracking. Enable fitness, gait analysis, and rehabilitation through real-time movement data.
Vision-Based Sensors	[193]	Utilize RGB, depth, and thermal imaging to recognize actions in complex scenes. Robust in static environments but sensitive to lighting and occlusion.
Physiological Sensors	[144]	Capture biosignals like EMG and ECG to monitor internal physical states. Enhance health-centric HAR by detecting muscle fatigue, stress, or abnormal vitals.
Environmental Sensors	[142]	Integrate microphones, pressure, and temperature sensors for ambient context awareness. Useful in smart homes for behavior prediction without user intervention.
Hybrid Sensors	[44]	Combine multiple sensing modalities (e.g., wearable + environmental) to boost recognition robustness. Facilitate multimodal fusion strategies for dynamic, real-world settings.

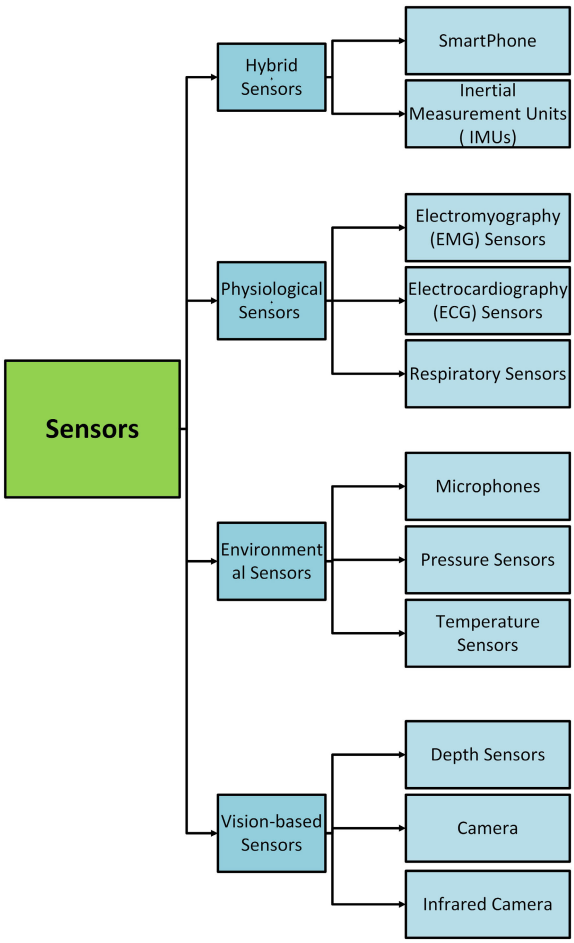

FIGURE 5. Taxonomy of sensor types used in HAR systems, categorized into hybrid, physiological, environmental, and vision-based sensors with representative modalities.

VII. DATASETS

High-quality datasets are essential for benchmarking and advancing HAR research. Over the past decade, a diverse range of datasets has enabled reproducible evaluation, fostered algorithmic innovation, and supported the development of generalizable recognition models [65], [166], [192]. However, persistent gaps remain in action diversity, contextual richness, and real-world representativeness [17].

Table 6 provides a comparative overview of widely used HAR datasets, focusing on their sensing modalities and distinctive attributes.

Trends and Unique Attributes:

- **Modality Coverage:** Data sets span wearables (UCI HAR, WISDM), multimodal sensor fusion (Opportunity, KU-HAR), environmental / ambient sensing (CASAS) and large-scale video (Kinetics, NTU RGB+D, HMDB51).
- **Scale and Diversity:** Modern datasets (e.g., Kinetics, CAPTURE-24) provide extensive action diversity and real-world variability, supporting DL and transfer learning.
- **Annotation Quality:** Recent efforts (e.g., CAPTURE-24) focus on free-living, long-duration, and hierarchical annotations, enabling open-set and few-shot recognition research [67].
- **Fusion and Context:** Opportunity and KU-HAR enable robust multimodal fusion, while CASAS and CAPTURE-24 emphasize context-rich and naturalistic settings.
- **Benchmarking Role:** UCI HAR, WISDM, and HMDB51 remain widely used for baseline benchmarking, but may lack complexity and generalizability for next-generation HAR.

Gaps and Limitations:

- **Action Diversity:** Many datasets focus on simple or scripted actions, limiting the evaluation of complex, overlapping, or spontaneous activities.

TABLE 6. Summary of major HAR datasets with their sensing modalities, key features, and distinctive characteristics.

Dataset	Sensing Modality	Key Features	Distinctive Attributes
UCI HAR [62]	Smartphone Accelerometers	Daily action benchmarking	Simple locomotion classes; widely used baseline for classical ML and DL
WISDM [162]	Wearables (Phones, Watches)	Fitness and ambulatory action monitoring	Focus on walking/jogging; supports low-complexity models; limited action diversity
Opportunity [61]	Multimodal (Body-worn + Ambient)	Context-rich indoor action monitoring	High-resolution, multi-location streams; supports hybrid models and sensor fusion
CASAS [160]	Environmental Sensors	Smart home and ambient sensing	Passive, non-intrusive; multi-resident tracking; strong for context-rich environments
PAMAP2 [162]	Inertial + Heart Rate Sensors	Health and rehabilitation monitoring	Fine-grained IMU and physiological data; supports energy/intensity-aware recognition
KU-HAR [204]	Multi-sensor Fusion (Wearables + Context)	Generic action classification	Robust to noise; suitable for real-world deployment and DL
Kinetics [215]	Video (RGB)	Large-scale AR	600+ action classes; 500,000+ YouTube clips; diverse, real-world video
NTU RGB+D [214]	RGB-D Cameras	3D AR	Large-scale, multi-view, skeleton and depth data; benchmark for DL
HMDB51 [66]	Video (RGB)	Realistic video actions	51 action classes; diverse sources (movies, web); challenging for temporal modeling
CAPTURE-24 [67]	Wrist-worn Accelerometers, Wearable Cameras	Free-living, long-duration action	3883 hours, 151 participants; naturalistic, annotated, supports open-set/zero-shot learning

- **Contextual Realism:** Laboratory-style datasets may not generalize to free-living or open-world scenarios; CAPTURE-24 and Kinetics address this gap.
- **Participant Diversity:** Few datasets include large, heterogeneous cohorts or multi-environment recordings.
- **Sensor Heterogeneity:** Most datasets focus on a single modality; only a few (Opportunity, KU-HAR, CAPTURE-24) support comprehensive multimodal fusion.

Future dataset development should prioritize real-world diversity, multi-modality, and annotation richness to support robust, generalizable HAR models.

VIII. CHALLENGES IN HAR

HAR systems encounter numerous obstacles that affect their scalability, functionality, and reliability. These challenges include safeguarding data confidentiality, ensuring sensor accuracy, and meeting the demand for instantaneous processing. Addressing these hurdles is essential for improving the effectiveness and broader implementation of HAR technologies across various sectors.

A. DATA CONFIDENTIALITY AND PROTECTION

Data confidentiality in HAR has become an urgent and technically complex challenge, especially in privacy-sensitive deployments such as eldercare monitoring, personalized healthcare, or smart home surveillance. These systems process real-time data streams—including motion patterns, ambient cues, and biometric signals—that are intimately tied to personal identity. As HAR expands beyond lab environments into wearable, environmental, and multi-user settings, the risks of data misuse, surveillance creep, and adversarial leakage intensify.

Protecting data confidentiality and ensuring robust security mechanisms remain significant challenges for HAR systems, particularly in sensitive domains such as healthcare and surveillance, where vast amounts of personal and sensitive information are processed. Managing the secure storage and transmission of data while safeguarding user privacy is a complex and non-negotiable requirement, often hindered by evolving technological and regulatory landscapes.

Despite numerous advances, several critical issues persist. For example, [39] emphasized the challenges of

implementing advanced encryption methods, secure communication protocols, and stringent access controls necessary to protect sensitive health data. Similarly, [33] discussed the complexities of developing a suitable security framework for healthcare environments enabled by 5G, highlighting the need for seamless integration with existing systems. [165] underscored the inherent difficulties in creating privacy-preserving methodologies that enable secure data sharing and processing, often constrained by the trade-off between performance and privacy.

In practical scenarios, challenges extend to applications such as fall detection, where maintaining user anonymity is critical. For example, [136] demonstrated the use of DL integrated with mmWave radar signals for privacy-focused fall detection, achieving notable accuracy but underscoring the difficulty of balancing efficacy and privacy. In surveillance, ethical data handling is paramount; [50] highlighted the ongoing struggle to establish effective anonymization practices, secure data repositories, and restrictive access mechanisms to prevent misuse of HAR data. Furthermore, managing sensitive information generated by IoT devices introduces additional vulnerabilities, as noted by [119], who explored frameworks to ensure compliance with privacy regulations while addressing potential security flaws.

Emerging privacy-preserving technologies such as federated learning, differential privacy, and homomorphic encryption now offer promising—but underutilized—solutions for secure HAR. In federated HAR settings, on-device models learn from local user data without transmitting it to centralized servers, reducing surveillance risks while enabling personalized adaptation. Blockchain-integrated HAR pipelines can also introduce tamper-proof audit trails for sensor data usage, supporting traceability and accountability in medical or surveillance contexts. Together, these paradigms are not futuristic luxuries—they are foundational to ethically scaling HAR systems across public and regulated domains.

Resolving these data privacy and security challenges requires the deployment of sophisticated cryptographic protocols and the adoption of highly secure data transmission techniques. Privacy-preserving machine learning algorithms must be further refined to address the vulnerabilities inherent in sensitive fields like healthcare and surveillance. Furthermore, implementing robust anonymization measures and enforcing stringent access control policies are essential to reduce the risk of breaches and unauthorized data exposure. The responsible and strategic use of IoT and 5G technologies poses another challenge, demanding that HAR systems strike a delicate balance between data protection and operational efficiency. Only by addressing these issues can HAR systems foster trust and achieve widespread adoption while adhering to privacy and security standards ([39]; [33]; [165]; [136]; [50]; [119]).

B. SENSOR ACCURACY AND CALIBRATION

Sensor accuracy and reliability are fundamental to the success of HAR systems. However, achieving and maintaining precise calibration presents a significant challenge. Issues such as sensor drift, variability in readings, and improper alignment can severely degrade system performance, necessitating advanced and adaptive calibration methods to mitigate these problems effectively. Stisen et al. [91] analyzed these discrepancies, highlighting the impact of mobile sensing heterogeneities on AR performance. Their findings emphasize the need for domain adaptation techniques and standardized calibration methods to improve the HAR model's robustness across diverse sensing platforms.

Wearable sensor-based HAR offers high granularity in movement tracking but struggles with sensor drift, limited contextual awareness, and high energy consumption. Mukhopadhyay [56] reviewed wearable HAR systems and noted that while they excel in motion-based classification, they often fail in scenarios requiring environmental context or object interactions. This highlights the importance of multimodal sensor fusion, where wearable sensors are complemented by vision-based or ambient sensing for a more holistic HAR system.

Sensor drift, a common issue over prolonged use, requires innovative solutions. For example, [143] demonstrated an averaging-based calibration method for smartphone microphones, improving environmental noise measurement accuracy. Wearable inertial sensors, like accelerometers and gyroscopes, are particularly prone to alignment errors and drift. As noted by [115], addressing these challenges demands precise alignment and robust drift correction techniques to maintain high system accuracy.

The persistent challenge of sensor calibration underscores the need for continuous advancements in self-correcting algorithms, adaptive sensor alignment methodologies, and domain-specific calibration frameworks to ensure that HAR systems perform reliably across diverse scenarios.

C. REAL-TIME PROCESSING

Real-time processing remains one of the most demanding requirements for HAR, particularly in critical domains such as healthcare, security, and predictive maintenance. The ability to analyze and respond to continuous data streams is crucial for timely interventions in fall detection, emergency response, and surveillance. Despite technological advancements, achieving low-latency inference without compromising accuracy remains a core challenge.

Latency minimization is a persistent bottleneck, particularly in systems processing multimodal sensor data. While edge computing and hardware acceleration have improved feasibility, real-time HAR demands further innovation in model design and computational efficiency. Conventional models such as CNNs and Transformers offer high accuracy but impose significant overhead, especially in resource-constrained environments.

Emerging paradigms like optical reservoir computing and neuromorphic hardware provide potential alternatives. Yang et al. [18] showcased the use of Semiconductor Optical

Amplifiers (SOA) for high-speed processing, indicating that non-conventional hardware may reduce inference time while conserving energy.

Latency can be formally expressed as:

$$\text{Latency} = \frac{\text{Processing Time}}{\text{Sensor Data Volume}}, \tag{17}$$

where optimization of both components is vital for real-time deployment.

Recent approaches emphasize lightweight networks, distributed processing, and quantization techniques such as CNN pruning to reduce system complexity [203]. While these methods show promise, integrating them with predictive capabilities and low-power requirements remains an open research challenge.

Predictive analytics can further enhance system responsiveness. For example, [37] applied predictive maintenance models to anticipate anomalies, highlighting the importance of forward-looking models in HAR. However, integrating such methods into time-critical applications adds complexity to system design.

Hardware acceleration remains a key enabler. FPGA-based solutions, as explored in [38] and [200], enable parallelized computation, which is essential for achieving sub-second inference. However, deployment of such architectures at scale requires significant development effort and cost considerations.

In summary, real-time HAR systems must balance latency, energy consumption, and predictive capacity. Continued research into hybrid model architectures and context-aware optimization is needed to meet the demands of dynamic environments.

D. ENERGY EFFICIENCY AND TEMPORAL DATA PREDICTION

Energy efficiency remains a critical constraint in real-time HAR, particularly for battery-powered devices like wearables and wireless sensors. Methods proposed by [32] and [182] aim to optimize energy consumption without degrading recognition accuracy. Wi-Fi-based HAR systems [53] exploit existing infrastructure to minimize energy costs, but these benefits often depend on environmental conditions and careful trade-offs with computational performance.

Temporal data prediction supports proactive responsiveness in HAR systems, especially in rapidly changing contexts. DL techniques, such as those from [35] and [176], enable accurate time-series modeling but introduce computational overhead. Balancing model complexity with hardware limitations remains a challenge in deploying these methods on lightweight platforms.

Future HAR frameworks must incorporate lightweight predictive models and context-aware energy optimization to ensure scalability and real-time performance across diverse deployment settings.

E. SENSOR INTEGRATION AND LATENCY REDUCTION

Efficient **sensor integration** is vital for minimizing delays in real-time HAR systems. In critical applications like assisted living, delays caused by poorly integrated sensors can severely compromise decision-making. [178] emphasized that optimizing multimodal sensor integration is essential for achieving rapid and dependable responses, but it requires sophisticated data synchronization and fusion techniques.

Edge and Fog Computing frameworks offer promising solutions to the latency challenges by decentralizing computational tasks. By processing data closer to the source, as demonstrated by [183], these frameworks reduce the reliance on centralized systems, enabling faster responses. However, implementing edge and fog computing for large-scale HAR deployments presents challenges in resource allocation and infrastructure design.

F. SUSTAINABILITY AND DOMAIN-SPECIFIC CHALLENGES

Ensuring the sustainability of HAR systems introduces unique obstacles, particularly in terms of energy consumption and environmental impact.

One of the key challenges in smart home-based HAR is recognizing actions in multi-resident environments, where overlapping actions and sensor ambiguity complicate classification. Mohamed et al. [51] provided a comprehensive literature review on multi-resident HAR, highlighting the limitations of existing models in distinguishing concurrent actions. Their findings emphasize the need for context-aware sensor fusion and personalized HAR models to improve recognition accuracy in shared living spaces.

Another major challenge is the reliance on predefined action labels, which limits the system's ability to generalize to unseen behaviors. Traditional HAR approaches focus on AR, where models classify predefined actions based on labeled training data. However, real-world human behavior is often dynamic and unstructured, necessitating action discovery—the process of identifying new, previously unknown actions from raw sensor data. Cook et al. [60] introduced the concept of combining action discovery with recognition, emphasizing that unsupervised and semi-supervised learning approaches can enhance HAR systems by enabling adaptation to novel user behaviors.

Despite significant advancements in DL-based HAR, several challenges remain. One major issue is the trade-off between model complexity and real-time performance, particularly in resource-constrained environments such as mobile and wearable devices. Nweke et al. [72] identified key challenges in DL-driven HAR, including high computational costs, the need for large-scale annotated datasets, and issues related to sensor heterogeneity and energy consumption. Addressing these challenges requires the development of lightweight DL models, efficient feature extraction techniques, and strategies for handling noisy and missing data in real-world scenarios.

Incorporating renewable energy sources has shown potential for addressing these concerns. For example, [31] explored renewable energy-powered HAR systems, demonstrating reduced dependency on conventional power sources while maintaining system efficiency. Nevertheless, integrating renewable energy technologies with HAR applications remains a challenging frontier.

Real-time HAR systems also face domain-specific obstacles, including:

- **Healthcare and Assisted Living:** Addressing sensor inaccuracies, minimizing latency, and mitigating environmental interference are critical for delivering consistent, real-time interventions ([178]).
- **Security and Surveillance:** Reducing false alarms and resolving processing delays are essential for ensuring effective threat detection in rapidly evolving scenarios.
- **Predictive Maintenance:** Extending device lifespan through predictive analytics necessitates efficient data integration and computational strategies, which remain challenging to implement at scale ([37]).

Key strategies and their impact on improving real-time HAR systems are summarized in Table 7.

Using advanced techniques such as hardware acceleration, predictive analytics, and decentralized computing, HAR systems can achieve reliable real-time processing, making them indispensable in critical real-world applications.

G. NOISE REDUCTION

One of the major challenges in HAR is ensuring robust recognition in noisy environments, particularly when using wearable sensors, acoustic signals, or RF-based HAR. Traditional noise filtering techniques, such as WTs and Kalman filtering, have been widely applied. However, recent advancements in phase-aware DL provide a new approach to noise resilience. Peer and Gerkmann [49] demonstrated that DL models leveraging phase information can significantly enhance speech signals in noisy environments. Similar techniques could be adapted for HAR, particularly in motion sensor data refinement and multimodal fusion, where signal distortions affect recognition accuracy.

Interference in sensor data is one of the major issues that affect the effectiveness of HAR systems, making it difficult to discern real signals of interest and leading to poor AR. Environmental interference and artifacts from the sensors themselves are some of the main causes of noise that degrade the quality of data. Noise reduction methods are vital to improve signal clarity and retention of relevant characteristics; however, optimization of these requirements remains a significant problem ([182]).

Sophisticated noise reduction techniques are central in helping reduce the effects of noise while preserving essential features. These methods are useful in the improvement of the dependability and precision of the HAR systems, especially in complex and ever-changing scenarios ([182]). Calibration methods are equally crucial when it comes to acoustic data. [143] developed an averaging method for calibrating smartphone microphones for assessing environmental noise with high precision to enhance the quality of the collected acoustic data and environmental monitoring. Likewise, [133] emphasized the need for calibration techniques for noisy settings while showing that their effectiveness can be beneficial to the HAR system in difficult conditions.

Reducing noise increases the reliability of the HAR system in all applications and provides consistent performance in noisy conditions while increasing the accuracy of AR. In healthcare, it strengthens the supervision of important actions such as fall detection. Sophisticated techniques of denoising and calibration facilitate accurate identification even under severe conditions.

H. DATA FUSION

Multimodal sensor data fusion is fundamental to HAR systems as it integrates heterogeneous sensor inputs—such as inertial, physiological, and environmental data—to construct a unified and context-rich understanding of human actions. While fusion enhances robustness and precision, it introduces challenges related to synchronization, computational overhead, and resource optimization ([184]).

To mitigate synchronization issues, [138] emphasized the role of UWB technologies for achieving accurate spatial and temporal alignment. Similarly, IoT-driven enhancements by [116] and [121] leverage real-time tracking via wearable sensors interconnected through adaptive fusion layers.

The computational intensity of fusion processes necessitates efficient frameworks. Strategies such as 5 G-enabled data pipelines ([36]) and cloud-edge integration architectures ([34]) have demonstrated reduced latency and improved scalability in HAR deployments. These infrastructures facilitate rapid aggregation of high-volume sensor streams, making them suitable for dynamic environments like smart cities.

Beyond conventional settings, fusion techniques extend HAR adaptability to unique domains. For instance, underwater HAR explored by [27] tackles signal degradation via novel calibration and sensor durability strategies. In smart home scenarios, frameworks such as [159] integrate multimodal inputs to support real-time, privacy-preserving action monitoring.

In sum, data fusion—when combined with synchronized sensing, scalable computing, and adaptive system design—enables HAR systems to operate reliably across complex and resource-variable environments.

I. POWER CONSUMPTION AND ENERGY EFFICIENCY

Wearable HAR devices are often required to deliver high performance while maintaining minimal energy consumption, especially in scenarios demanding continuous operation. Meeting these dual demands requires advances in both sensing technologies and energy-aware system design.

Cohn et al. [5] and Kim et al. [127] introduced the use of ultra-low-power sensors that rely on static electric field sensing, enabling continuous AR with minimal energy

TABLE 7. Overview of optimization techniques used to improve latency and efficiency in real-time HAR systems.

Method	Impact on Latency	Impact on Efficiency
Pruned CNN Layers	Reduces latency by simplifying network architectures and removing redundant parameters.	Maintains model accuracy while significantly lowering computational complexity.
Feature Quantization	Speeds up processing by reducing data precision and bit-depth representation.	Lowers memory and resource usage, making models more energy-efficient.
Hardware Acceleration (e.g., FPGA)	Provides real-time processing capabilities with optimized hardware circuits.	Increases computational throughput, improving speed and power efficiency.
Edge/Fog Computing	Reduces network latency by performing computations closer to the data source.	Minimizes cloud server dependency, enhancing real-time system responsiveness.
Energy-Efficient Models	Optimizes model architectures for minimal power consumption in embedded systems.	Extends battery life, making HAR systems suitable for long-term wearable applications.

overhead. Liu et al. [191] demonstrated that real-time energy allocation using machine learning techniques allows systems to dynamically change power usage according to workload, thus preventing unnecessary battery drain.

Key strategies include optimizing sensor HARdware for low power draw, developing power-aware learning algorithms, and implementing HARdware-level energy design. These approaches collectively ensure that wearable HAR systems maintain stable performance and energy efficiency during continuous motion tracking in real-time applications.

These challenges are also summarized in Table 8 below:

Table 8 illustrates how certain challenges are more pronounced in specific domains. For instance, sensor calibration issues are highly relevant in smartphone-based or wearable HAR but less significant in passive systems like ambient sensors. Conversely, data privacy and surveillance risks are elevated in vision-based HAR, especially under 5G healthcare infrastructures. Similarly, predictive analytics and energy efficiency are critical for edge-based and wearable deployments, whereas cloud-centric architectures prioritize resource allocation and data fusion. These distinctions highlight that the selection of HAR strategies must be informed by domain-specific constraints and technical trade-offs.

IX. APPLICATIONS

HAR has become increasingly relevant across various domains, including consumer healthcare, smart environments, smart cities, and sports. By integrating sensor technologies with advanced computational models, HAR improves decision-making, improves safety, and allows personalized user experiences. Beyond traditional sensor-based approaches, HAR systems benefit from context-aware modeling, where environmental and situational factors contribute to improved classification accuracy. For example, Liu et al. [74] proposed a dynamic embedding model for context-aware HAR, demonstrating its adaptability to varying action patterns based on contextual dependencies. Their findings highlight the importance of incorporating contextual information to improve recognition accuracy and ensure robust performance in diverse real-world settings.

This section explores the applications of HAR technologies, focusing on their computational demands, deployment challenges, and integration into real-world scenarios.

A. HAR IN SURVEILLANCE AND CROWD ANALYSIS

HAR is widely applied in surveillance and crowd analysis, where recognizing interactions and collective behaviors is crucial for security and public safety. Multi-modal approaches enable HAR systems to capture both individual and group actions, improving event detection in complex environments. Li et al. [98] introduced a method for group AR by learning multi-modal densities on a discriminative temporal interaction manifold, demonstrating its effectiveness in identifying structured interactions within crowds. Recognizing group actions requires analyzing spatial relationships and interaction patterns among individuals. Cho et al. [99] proposed a group AR approach based on interaction zones, leveraging relative distances between human objects to improve classification accuracy. Their findings highlight the significance of spatial modeling in HAR for surveillance applications, where understanding group dynamics is essential for behavior analysis and security monitoring.

B. HEALTHCARE MONITORING

Automated home health monitoring and rehabilitation analysis systems, commonly known as HAR systems, play a vital role in real-time patient monitoring and early disease detection, particularly in critical applications such as fall prevention and rehabilitation. These systems must balance high accuracy with low latency to ensure reliable performance.

Verbunt et al. [103] examined methods for assessing physical action in patients with musculoskeletal pain, highlighting the role of continuous monitoring in understanding mobility

TABLE 8. Multidimensional overview of technical challenges and application contexts in HAR.

Focus Area	Application Context	Key Challenges Addressed
Data Privacy and Security	Secure data transmission [39]; Surveillance systems [50]	Advanced encryption mechanisms; Access control and secure storage
Secure 5G-Enabled HAR	Healthcare IoT networks [33]	Real-time threat mitigation; Secure, low-latency transmission
Privacy-Preserving HAR	Fall detection via mmWave radar [136]	Anonymous sensing techniques; Non-visual data collection
Sensor Calibration	Smartphone microphones [143]; Pressure-based balance measurement [29]	Environmental calibration accuracy; Standardized baseline compensation
Predictive Analytics	Real-time maintenance systems [37]; Time-series action prediction [35]	Temporal modeling under uncertainty; Action anticipation for early response
Energy Efficiency	Wireless sensor networks [32]; Low-power wearables [158]	Power-aware sensing; Static electric field harvesting
Noise Reduction	Acoustic calibration (smartphones) [143]	Background denoising; Signal enhancement under clutter
Resource Allocation	5G network optimization [36]	Latency-budget scheduling; Bandwidth-efficient HAR processing
Data Fusion	Multimodal HAR architectures [184]	Sensor synchronization; Fusion latency mitigation
UWB Technology	Precision localization [138]	Short-range high-fidelity tracking; Integration with wearable HAR
IoT and Cloud Computing	Real-time HAR monitoring [34]	Cloud-edge offloading; Real-time inference constraints
Underwater HAR	Submerged signal processing [27]	Environmental resilience; Sensor durability challenges

patterns and treatment outcomes. Similarly, Bian et al. [28] developed a wearable magnetic field proximity sensor for controlling social contact, demonstrating the adaptability of HAR systems to diverse public health challenges. Liu et al. [148] integrated wearable ECG and accelerometer data, emphasizing the importance of efficiently fusing physiological and motion signals for real-time monitoring in resource-constrained environments.

In the context of smart healthcare environments, Subasi et al. [180] explored machine learning methods for HAR, demonstrating their potential in enhancing predictive analytics and personalized patient care. Their findings highlight the growing role of AI-driven HAR systems in improving diagnostic accuracy and optimizing treatment plans.

For rehabilitation applications, Homayounfar and Andrew [118] reviewed wearable sensor technologies, discussing their underlying mechanisms, material innovations, and challenges in motion monitoring. Badwaj et al. [141] studied the impact of limb movement on UWB antenna performance, focusing on optimized hardware designs for accurate data transmission during dynamic actions.

On a broader scale, Oguntala et al. [104] introduced SmartWall, an RFID-enabled ambient HAR system that eliminates wearables by offloading computation to centralized servers, offering a scalable solution to monitor vulnerable populations. Hossen et al. [107] developed a mobile automated medical assistant designed for hospitals, showcasing the potential of AI and IoT-driven HAR applications in healthcare environments. Amft et al. [24] explored the role of wearables in symptom tracking and contact tracing, highlighting their potential in early disease detection and outbreak control.

Despite these advancements, challenges such as integrating diverse sensor modalities and managing large-scale healthcare datasets persist. Addressing these issues will require the adoption of edge computing and lightweight machine learning algorithms to overcome computational bottlenecks and improve scalability.

C. SMART HOMES AND CITIES

HAR systems in smart homes enhance safety and personalization by analyzing actions and adapting the environment accordingly. Bangaru et al. [145] proposed Artificial Neural Network (ANN)-based HAR framework for scaffold builder AR using wearable EMG and IMU sensors, demonstrating its effectiveness in classifying construction-related tasks. These systems face computational challenges, including real-time processing and efficient resource utilization. Non-intrusive solutions, such as UWB radar-based systems proposed by [139], enable action detection but require significant computational power for signal processing. Similarly, [120] introduced device-free AR using WiFi, eliminating the need

for wearables while addressing noisy and large datasets. Ramos et al. [42] demonstrated that daily human AR can be effectively achieved using non-intrusive sensors. Their study highlights the potential of ambient intelligence in smart homes, reducing the need for wearable devices while ensuring accurate motion detection. Jalal et al. [96] proposed a HAR approach utilizing recognized body parts from human depth silhouettes for resident monitoring in smart homes. Kashimoto et al. [25] proposed ALPAS, an analog PIR sensor-based AR system that enables real-time motion detection in smart homes. In the context of smart cities, HAR systems contribute to safety and mobility, as demonstrated by [137], who integrated WiFi fingerprinting with random forest algorithms to achieve accurate indoor positioning—an essential component of urban mobility. However, scaling such systems requires robust data fusion frameworks and substantial computational resources. To address these challenges, edge AI is critical for enabling distributed processing and minimizing latency, ensuring the viability of real-time applications in both urban and residential settings.

D. FITNESS AND SPORTS

HAR systems play a crucial role in fitness and sports by enabling real-time motion tracking and response. For instance, [131] demonstrated that IMU-based systems can achieve high accuracy in exercise recognition, though lightweight models are necessary for battery-powered wearables. Similarly, [27] explored capacitive sensing technologies for gesture tracking, emphasizing the need for real-time feature extraction and classification. These systems also help prevent athletic injuries by providing athletes with valuable insights about their bodies, though effective preprocessing and real-time feedback remain essential for user safety and satisfaction. Posture-based HAR methods utilize spatial and temporal information from skeletal or silhouette data to model action transitions effectively. Reyes-Ortiz et al. [181] proposed a transition-aware HAR framework using smartphone sensors, demonstrating improved recognition accuracy by modeling transitions between actions. Kellokumpu et al. [122] proposed a HAR framework based on sequences of postures, demonstrating its effectiveness in recognizing structured human movements. The integration of IoT with HAR enables scalable and adaptive AR, facilitating smart home automation, elderly care, and security monitoring. Diraco et al. [186] explored the role of IoT in HAR, emphasizing the importance of sensor interoperability and energy-efficient processing in smart environments. The integration of multiple sensors further enhances HAR accuracy and reliability but increases computational complexity, as seen by [169], where LoRa RF signals were combined with inertial data for improved recognition in long-range IoT applications, and by [95], where WTs were used to extract temporal and frequency features for transient action analysis. Addressing this complexity requires efficient multimodal architectures and optimized feature extraction techniques. Case studies highlight both the potential and challenges of HAR systems; for example, [104] demonstrated ambient monitoring in healthcare but required centralized computing, while smartphone-based HAR systems [97] have proven effective for urban navigation but face issues like power consumption and scalability. Emerging approaches such as edge computing and federated learning, distribute computational tasks to improve scalability, while quantized neural networks enable HAR systems to function on resource-limited devices, optimizing power consumption without significantly sacrificing accuracy.

Table 9 reveals that the design and deployment of HAR systems vary significantly across applications. For instance, in healthcare monitoring, prioritizing real-time responsiveness and privacy mandates lightweight, secure models. In contrast, smart home HAR emphasizes environmental robustness and energy efficiency, often requiring radar or ambient sensors instead of wearables. Meanwhile, fitness-focused HAR prefers wearable-based systems optimized for power efficiency and motion precision. Sensor fusion, while enhancing recognition accuracy, introduces complexity in synchronization and processing overhead, making it more suitable for high-resource environments like smart cities than for edge devices. These comparisons clarify how application-specific constraints influence architectural choices in HAR system design.

In many areas, HAR systems can improve performance and demonstrate innovative applications using new models and sensor integration. However, solving issues related to scalability, resources, and real-time processing can still be important for achieving broader applications.

Identifying challenges is only part of the equation; it is equally important to quantify a system's ability to address them. A robust evaluation framework ensures that HAR models are not only accurate but also efficient, scalable, and resilient to environmental variation. The following section introduces a comprehensive set of metrics designed to holistically assess HAR systems across these dimensions.

As HAR systems become increasingly embedded in real-world environments, it is no longer sufficient to rely solely on raw sensor signals for accurate recognition. Context-awareness is vital—for instance, understanding whether a person is running outdoors vs. indoors, or falling in a crowded room vs. an empty hallway, requires more than motion data. Emerging models incorporate contextual cues using knowledge graphs, location embeddings, and semantic reasoning frameworks that fuse sensor data with time, space, social context, and user identity. Such models not only improve accuracy but also reduce false positives in ambiguous settings and enable behavior prediction in healthcare, surveillance, and assistive systems.

X. EVALUATION METRICS

The evaluation of' HAR systems must consider key performance indicators such as accuracy, computational efficiency, robustness, and scalability to ensure real-world applicability.

TABLE 9. Application-specific computational requirements and deployment challenges in HAR systems.

Application Area	Computational Insights and Deployment Challenges
Healthcare Monitoring [104], [141]	Requires low-latency algorithms for continuous patient observation. Major challenges include maintaining diagnostic accuracy, ensuring reliable network connectivity, and addressing patient data privacy.
Smart Homes and Cities [120], [139]	Utilizes device-free HAR via WiFi/radar signals. Balances large-scale deployment with real-time processing. Faces challenges such as signal occlusion, dynamic environments, and energy constraints.
Fitness and Sports [27], [131]	Demands real-time motion capture using compact wearable sensors. Limitations include short battery life, sensor drift during intense motion, and the need for lightweight yet accurate models.
Multimodal Sensor Fusion [169]	Enhances HAR robustness through cross-modal signal integration. Challenges involve synchronizing asynchronous sensor streams, managing computation across modalities, and ensuring fusion efficiency.

Traditional metrics like accuracy, precision, recall, and F1-score remain fundamental, but with increasing real-time constraints, factors such as energy efficiency and processing latency have become critical. Bustoni et al. [48] analyzed classification models for HAR, demonstrating that DL methods outperform traditional classifiers such as SVM and Random Forest in complex tasks. However, their findings also highlight computational trade-offs, where increased accuracy often comes at the cost of higher processing demands. This necessitates evaluating models not only on accuracy but also on computational efficiency, noise resilience, and real-time adaptability. Therefore, the key evaluation metrics should be the following:

A. ACCURACY

Accuracy is the simplest measure and means the capacity of the model to correctly predict actions out of the total number of possible actions ([92]). Accuracy gives a general idea about the performance and is misleading if the distribution is skewed; a model can be accurate but is simply memorizing the data from the most frequent classes (see Equation 18).

$$\text{Accuracy} = \frac{\sum \text{Correct}}{\sum \text{Predicted}}, \tag{18}$$

B. PRECISION, RECALL, AND F1-SCORE

The **Precision** describes the ratio of TP to the total number of positive predictions, taking a closer look at the quality of the positive predictions ([90]). This is especially important in areas such as medicine where the identification of false positives may necessitate the undertaking of other invasive tests (see Equation 19).

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}, \tag{19}$$

Continuing with the next two, accuracy, or recall, measures the ratio of true positives relative to total actual positives, and thus is important for identifying vital actions such as falls in a healthcare setting ([89]) (see Equation 20).

$$\text{Recall} = \frac{\text{No. of true +ve}}{\text{No. of true +ve} + \text{No. of false-ve}}, \tag{20}$$

The **F1-Score** is between Precision and Recall, and thus, for setups with imbalanced classes, this is an adequate measure ([84]) (see Equation 21).

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{21}$$

C. CONFUSION MATRIX

The confusion matrix provides a detailed breakdown of model performance by visualizing true positive, false positive, true negative, and false negative predictions ([89]). It helps pinpoint areas where the model excels or needs improvement.

TABLE 10. Standard confusion matrix representation used for evaluating HAR system performance.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

D. AREA UNDER THE ROC CURVE (AUC-ROC)

The AUC–ROC evaluates the model's ability to distinguish between classes. The ROC curve plots the true positive rate (Recall) against the false positive rate (1–specificity) [85]. A higher AUC indicates better classification performance.

E. SCALABILITY METRICS

Modern HAR systems must handle increasing sensor data volumes while maintaining performance. Scalability can be evaluated using the following metrics:

- **Throughput:** The number of actions processed per second as sensor data volume increases.
- **Model Degradation Rate:** The percentage decrease in accuracy or latency when sensor data volume doubles.

F. ENERGY EFFICIENCY

Given the power constraints of wearable HAR systems, energy efficiency is a key performance metric. Reflects the system's ability to perform recognition tasks without excessive power draw, which is essential for long-term deployment and user convenience. The following indicators quantify energy-related performance:

- **Energy Consumption:** Average energy (in Joules) required per AR task, given by (see Equation 22):

$$\text{Energy Con} = \frac{\text{Tot Power Usage}}{\text{No. of Actions Processed}}, \tag{22}$$

- **Battery Impact:** Estimated percentage reduction in device battery life after continuous HAR operation over a fixed time interval.

G. ROBUSTNESS TO NOISE

HAR systems deployed in real-world environments must handle sensor noise and environmental variability. Robustness to noise can be quantified using:

- **Noise Tolerance:** The maximum signal-to-noise ratio (SNR) at which the system maintains a predefined accuracy threshold.
- **Accuracy Degradation:** The percentage decrease in accuracy when noise levels are increased, measured in various environments.

H. REAL-TIME PERFORMANCE METRICS

As for the real-time usage of HAR to, for example, monitor health or control surveillance systems, low latency and optimization of the computational process are crucial [86].

- **Inference Latency:** The average time (in milliseconds) taken to compute an action when the sensor data are received ([88]).
- **Computational Load:** The mean of the CPU and memory consumption during the execution of the model ([87]).

I. LINKING METRICS TO REAL-WORLD APPLICATIONS

- **Healthcare:** Low latency is crucial for fall detection and emergency interventions. Noise robustness ensures accurate monitoring in dynamic environments like hospitals.
- **Wearables:** Energy efficiency directly impacts user experience, as longer battery life enhances the usability of the device.
- **Smart Cities:** Scalability metrics make it possible to determine whether HAR systems can support large-scale deployment that incorporates multiple sensor networks.

Because modern HAR challenges include scalability, energy efficiency, and noise resistance, the proposed framework includes a set of evaluation measures that can cover all aspects of the HAR application.

Having examined the performance metrics and technical capabilities of modern HAR systems, it becomes essential to reflect on the broader trajectory of this domain. The following section synthesizes key findings, reiterates unresolved limitations, and outlines forward-looking directions to enhance HAR applicability, scalability, and ethical deployment.

While deployment-aware metrics like latency and energy efficiency are increasingly recognized, most HAR studies still emphasize classification accuracy as the primary benchmark. This narrow focus risks overlooking crucial aspects such as cross-user fairness, model bias, resilience under missing data, and generalizability across sensor placements or devices. A comprehensive evaluation framework should include: (1) latency (ms), (2) power usage (mW), (3) demographic fairness (across age/gender/mobility), (4) robustness to sensor dropout, and (5) on-device adaptability. Incorporating these factors will ensure HAR systems are not only accurate but also responsible, efficient, and scalable in real-world settings.

XI. CONCLUSION AND FUTURE WORK

This review systematically evaluates the advances in HAR by analyzing sensor technologies, signal processing methods, and machine learning algorithms. The contributions of this work include:

- **Advancements in Sensor Technologies:** HAR has benefited significantly from the integration of capacitive sensors and 3D magnetic sensors, enabling high-precision recognition of hand gestures and motion. [20] explored real-time hand gesture recognition, while [101] highlighted its applications in human-robot interaction, linking HAR with adaptive robotics.
- **Multimodal Sensor Fusion:** The combination of accelerometers, gyroscopes, EMG, and vision-based sensors enhances HAR accuracy, particularly for actions involving complex motion and environmental interaction.
- **Real-World Applications:** HAR systems have shown transformative potential across multiple domains, including healthcare, smart homes, fitness, surveillance, and urban planning.

Despite these advancements, HAR still faces persistent challenges, including noise susceptibility, real-time data processing limitations, scalability issues, and data privacy and security concerns. Additionally, the absence of standardized evaluation metrics and validation protocols hinders the comparability and reproducibility of HAR systems across different applications and datasets.

A. FUTURE WORK

To move beyond the limitations of prior HAR surveys and address pressing real-world needs, this review outlines a forward-looking roadmap built around emerging research themes and deployment-centric challenges. Unlike earlier surveys that focused predominantly on CNN and RNN architectures, this work emphasizes the convergence of classical signal processing with next-generation models—particularly transformers, graph-based networks, adaptive fusion, and privacy-aware edge architectures. Our review uniquely anchors future HAR directions in deployment

feasibility, energy efficiency, interpretability, and ethical AI—positioning itself as a bridge between algorithmic advancement and real-world applicability.

- **Next-Generation HAR with Transformer Architectures:** Vision Transformers (ViTs), spatio-temporal attention models, and HAR-specific transformer variants outperform CNN/LSTM baselines in long-range dependency modeling and multimodal fusion. Their ability to operate across asynchronous sensor inputs makes them promising candidates for robust recognition in noisy and overlapping action scenarios.
- **IoT-Driven Federated Learning for Privacy and Scalability:** Distributed HAR architectures using federated learning allow model updates without data centralization. This enhances privacy and scalability across heterogeneous devices (wearables, cameras, microphones) while enabling localized personalization without compromising data protection.
- **HARdware-Aware Optimization and Energy-Efficient Architectures:** Real-world deployment of HAR models requires techniques like dynamic pruning, quantization-aware training, and mobile-specific design (e.g., TinyML, MobileNet). These strategies enable inference on edge devices under strict energy, memory, and latency budgets.
- **Multimodal Adaptive Sensor Fusion:** Static early/late fusion paradigms fall short in dynamic settings. Future systems must adopt adaptive fusion using attention mechanisms, reinforcement learning, and graph-based modeling to contextually weigh input modalities during inference, especially under sensor drift or partial failure.
- **Privacy-Preserving HAR with Blockchain and Encryption:** As HAR systems move into healthcare, surveillance, and public settings, trustworthiness becomes paramount. Techniques such as homomorphic encryption, differential privacy, and blockchain-based audit trails must be embedded into HAR pipelines to ensure transparency and prevent data misuse.
- **Context-Aware and Situation-Responsive Models:** Future HAR systems should reason beyond raw sensor data. Embedding semantic context (e.g., time of day, user role, object proximity) via knowledge graphs or embeddings will enable more nuanced decision-making and reduce false positives in ambiguous environments.
- **Benchmarking Protocols Beyond Accuracy:** Accuracy alone is no longer sufficient. Standardized evaluation must include robustness to noise, cross-device generalizability, fairness (across demographics), latency (in ms), and energy consumption (in mW). We recommend a shift toward multidimensional benchmarking.
- **Large-Scale Real-World Deployments:** The final frontier of HAR research lies in translation. Large-scale, privacy-compliant deployments in hospitals, smart homes, and cities must be pursued. Edge devices tested under real-world constraints (bandwidth, occlusion, user variability) will ultimately validate model robustness.

In essence, this review differentiates itself by synthesizing foundational HAR models with a focus on scalable deployment, ethical compliance, and algorithmic innovation. By advocating for transformer-based models, neuromorphic strategies, and adaptive fusion, we position this survey as a next-generation reference for both academic and applied HAR research.

LIST OF ABBREVIATIONS

AI	Artificial Intelligence.
ANN	Artificial Neural Network.
AR	Action Recognition.
CNN	Convolutional Neural Network.
DL	Deep Learning.
EMG	Electromyography.
FT	Fourier Transform.
GCN	Graph Convolutional Network.
HAR	Human Action Recognition.
HCI	Human-Computer Interaction.
LSTM	Long Short-Term Memory.
PCA	Principal Component Analysis.
RFID	Radio-Frequency Identification.
RNN	Recurrent Neural Network.
UWB	Ultra-Wideband.
WT	Wavelet Transform.

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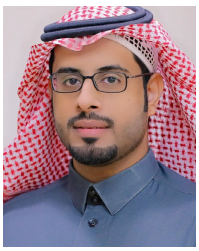

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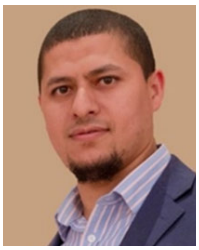

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